Agents guide

How an agent identifies itself, which machines may take a job, and how it signs in.

The agent calls the control plane

The control plane never dials the host. The agent process opens HTTPS to the server, registers under its own name, sends a heartbeat, and asks for work. The name is the routing key. The hostname is a label for people.

Example, private network, local operator login:

java -jar automation-agent.jar --server https://control.example.com --name finance-host-01 --hostname finance-01.corp --user operator --password <secret>

Example, remote agent, short-lived token:

java -jar automation-agent.jar --server https://control.example.com --name finance-host-01 --client-id justrunit-agent --client-secret <secret> --token-url https://login.example.com/oauth/token

The server address and the token address must be HTTPS unless the host is localhost. The agent does not use a person's password in this mode. Environment names for the token path are AUTOMATION_CLIENT_ID, AUTOMATION_CLIENT_SECRET, and AUTOMATION_TOKEN_URL.

Name and hostname

Field	Role
name	Unique identity. Used for register, heartbeat, claim, and preferred agent.
hostname	Optional label, up to 256 characters. Not used for networking or for choosing who runs a job.
capacity	How many claims the control plane will hand this name before it waits.
load	The count shown in the UI is the work actually running, not a stale counter.
process	One agent process runs one job, then asks for the next. Run several processes on a host when several jobs must run at the same time.

Hostname comes from the --hostname flag, otherwise the HOSTNAME environment variable, otherwise localhost. Registering the same name again reconnects that agent and updates the label, capacity, and heartbeat. A different name is a different agent.

Pool and preferred agent

A job belongs to a workspace. The workspace may name an agent pool. The job may also name one preferred agent.

Pool	Preferred agent	Who takes the next run
None	Empty	Any online agent with free capacity.
Set	Empty	Any online member of that pool.
Either	One name	Only that agent, even if it is outside the pool. The job waits if that host is down or full.
Empty pool	Empty	Nobody. The job stays queued until agents join the pool.

Set the preferred agent to the agent name, not the hostname, unless you also named the agent that way. Members of a pool need the same scripts and paths, because either online member may take the next job. Adding or removing a member changes who may claim. It does not move a run that is already in progress.

The account on the host

On Linux the agent refuses to start as root. The command runs in the shell of that process, so it has only that account’s rights. There is no setting to run one job as another user. Two jobs that need different rights use two agents, usually in two pools. The packaged containers drop every privilege and keep the root filesystem read-only. The job can write its temporary directory and its artifact cache.

A command succeeds when the process exits 0. Linux uses /bin/sh -c. Windows uses cmd.exe /c.

What the control plane stores

The agent row keeps the name, the hostname label, online or offline, capacity, and the last heartbeat. A missed heartbeat marks the agent offline. A run already in progress is then marked orphaned. An operator requeues it after confirming the process on that host is gone. Another pool member can take the requeued run.

Action	Path	Keyed by
Register	POST /api/v1/agents/register	name
Heartbeat	POST /api/v1/agents/{name}/heartbeat	name
Claim	POST /api/v1/agent/{name}/claim	name
Complete	POST /api/v1/agent/{name}/complete/{executionId}	name
Remove	Operator or administrator in the UI, or the agents API	name

Register and remove require an operator. A viewer can see the agent list and cannot register or remove an agent. The same name on a later start is a reconnect, not a second agent. If the process starts under a different name, the control plane treats it as a different agent and the first name goes offline after the heartbeat is missed.

There is no DNS lookup and no connection from the control plane to the hostname.